# Memory Is the Price: Evidence from the LLM Serving Market on the Economics of Mixture-of-Experts Inference

(Extended Abstract)

Michael J. Yuan (ByteFuture Inc.)     Ju Long (Texas State University)

**Abstract.** Sparse Mixture-of-Experts (MoE) models are widely claimed to cost what their *active* compute costs. We test this claim against the LLM serving market itself. Because providers disclose almost nothing about their production systems, we treat posted prices as an observable proxy for the latent serving configuration. A hedonic regression over 121 provider–model price observations rejects the claim: the market bills MoE models most of the way toward their total memory footprint (sparsity elasticity 0.554, $t = 8.3$). Market structure explains why. Capturing MoE's compute savings requires expert-parallel serving pools far larger than one node. Price floors therefore belong exclusively to vertically integrated neoclouds, hyperscalers list identical models at 2–5× the floor, and provider participation collapses with model footprint.

## 1 Motivation: an efficiency claim the market can test

Frontier open-weight language models have shifted decisively to sparse MoE architectures, which store hundreds of billions of parameters but activate only a small fraction per token: DeepSeek-R1 activates 37B of 671B, and Kimi K2 activates 32B of one trillion. The architecture's economic pitch is that customers pay only for active compute. Switch Transformers promised "outrageous numbers of parameters" at "a constant computational cost" [1], and Mistral stated that Mixtral "processes input and generates output at the same speed and for the same cost" as a dense model the size of its active parameters [2].

Whether the market delivers this discount is difficult to check directly, because commercial inference is an unusually opaque industry. Providers publish per-token prices, but almost never the cluster scale, parallelism, served precision, batch size, utilization, or margin of any endpoint, and black-box audits show that 11 of 31 commercial Llama endpoints served output distributions deviating from the reference weights [3]. The serving configuration is thus a latent variable, and the posted price is its observable proxy. Open-weight serving approximates the textbook competitive case: the weights are free, the software is open source, entry is unrestricted, and dozens of providers quote side by side on the OpenRouter marketplace. Competition therefore disciplines prices toward marginal cost, and sustained price structure identifies cost structure [4].

# 2 Hypotheses

Our empirical strategy follows the hedonic-pricing tradition [5]: we regress prices on the characteristics that drive resource consumption and read the coefficients as the market's implicit prices for each resource. In

$$\log(\text{price}) = c_0 + c_1 \log(\text{active\_params}) + c_2 \log(\text{total/active}), \tag{1}$$

$c_1$ values active compute and $c_2$ values resident-but-inactive capacity. **H1 (compute pricing, the industry claim)** predicts $c_2 = 0$; if memory rather than compute is the binding cost, $c_2 \approx c_1$. **H2 (scale economies)** holds that the minimum efficient scale of large MoE serving exceeds a single node, because the compute savings are only captured when traffic from thousands of users is aggregated across expert-parallel decoding pools [6, 7]. H2 predicts that price floors are set by massively parallel "neocloud" providers, that general-purpose clouds price above them, and that providers who cannot justify such pools exit rather than list, which compresses observed price dispersion through selection [8].

## 3 Data and method

We collected per-provider prices for 27 candidate open-weight models from OpenRouter's public endpoints API on 2026-07-01, cross-checked against provider price pages. After exclusions, the panel comprises 121 observations across 18 models (12 MoE, 6 dense) from roughly 40 providers, joined to architecture parameters verified against each model's HuggingFace configuration file. All statistics are computed deterministically; the data and code are public.[1]

## 4 Results

**H1 is rejected.** The sparsity ratio, which compute pricing predicts to be free, is priced at 83% of the rate of active compute (Table 1). A 671B-total/37B-active model is billed like a $\sim$430B dense model, and every MoE model sits 2.4–9.3$\times$ above the dense-only price schedule (Figure 1). Customers keep almost none of the advertised discount.

Table 1: The table reports OLS estimates of regression (1) ($n = 18$ models, $R^2 = 0.866$). H1 predicts $c_2 = 0$; the measured ratio is $c_2/c_1 = 0.83$.

| Coefficient | Estimate | Std. err. | $t$ |
|---|---|---|---|
| $c_1$: log(active parameters) | $+0.665$ | 0.099 | 6.7 |
| $c_2$: log(total/active) | $+0.554$ | 0.067 | 8.3 |

**H2 is supported.** Published deployments show a $\sim$36$\times$ per-GPU decode throughput gap between a 96-GPU expert-parallel pool and a single 8-GPU node, so no provider can sell at the market floor from a single-node deployment [7, 9]. In all seven panels for models of 200B+ total parameters, the floor is set by a vertically integrated neocloud, six of seven by DeepInfra alone. Hyperscalers list identical models at 2–5$\times$ the floor. Participation collapses with footprint, from 17 providers for a 117B-parameter model to three for a one-trillion-parameter model. A corollary

[1] https://github.com/juntao/memory_is_the_price

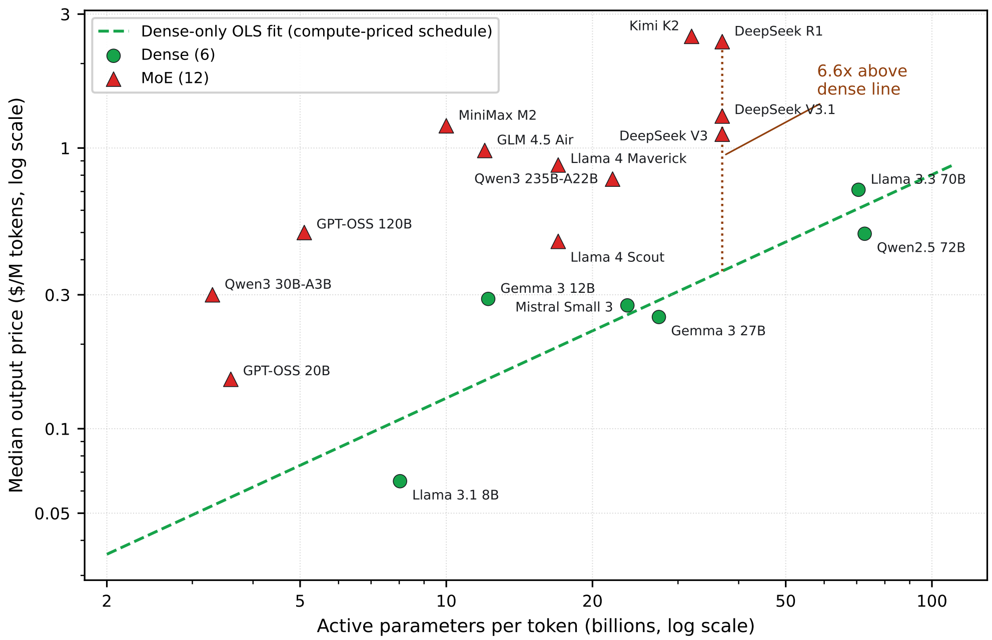


Figure 1: The figure plots median output price against active parameters for all 18 models on log-log axes. Every MoE model (triangle) sits above the dense-only OLS fit (dashed line), which is the schedule a compute-priced market would apply.

result is that MoE price dispersion is *compressed* rather than widened (median coefficient of variation 0.34 versus 0.54 for dense): providers whose latent cost exceeds the competitive price do not list at all, so the observed prices are a selected sample [8].

# 5 Contribution and implications

To our knowledge this paper provides the first hedonic estimate of the sparsity elasticity of price, formalizing an observation made informally on five models by Scher [10] and raised descriptively by Li et al. [11]. It also provides the first market-structure evidence that large-MoE serving is gated by scale. For information-systems and economics audiences, the study demonstrates a general method: using marketplace prices as revealed cost structure for digital infrastructure whose production technology is deliberately undisclosed. For practitioners, the results imply that MoE serving economics are governed by memory rather than FLOPS, and that the economically rational serving hardware outside hyperscale pools inverts the GPU design point: large commodity memory, modest compute, and moderate bandwidth.

## References


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